

One Table, Two Study Designs

AP Statistics · Unit 3 · Topic 3.14 · B: 2026-12-22 · E: 2027-02-05 · TEACHER KEY

CED TETHER

- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-8** — The chi-square distribution may be used to model variation.
- **LO VAR-8.I** — Identify the null and alternative hypotheses for a chi-square test for homogeneity or independence.
- **EK VAR-8.I.1** — The appropriate hypotheses for a chi-square test for homogeneity are: H_0 : There is no difference in distributions of a categorical variable across populations or treatments. H_a : There is a difference in distributions of a categorical variable across populations or treatments.
- **EK VAR-8.I.2** — The appropriate hypotheses for a chi-square test for independence are: H_0 : There is no association between two categorical variables in a given population (or the two variables are independent). H_a : Two categorical variables in a population are associated or dependent.
- **LO VAR-8.J** — Identify an appropriate testing method for comparing distributions in two-way tables of categorical data.
- **EK VAR-8.J.1** — When comparing distributions to determine whether proportions in each category for categorical data collected from different populations are the same, the appropriate test is the chi-square test for homogeneity.
- **EK VAR-8.J.2** — To determine whether row and column variables in a two-way table of categorical data might be associated in the population from which the data were sampled, the appropriate test is the chi-square test for independence.
- **LO VAR-8.K** — Verify the conditions for making statistical inferences when testing a chi-square distribution for independence or homogeneity.
- **EK VAR-8.K.1** — Conditions for chi-square tests on two-way tables (homogeneity or independence): a. Independence check: i. For a test for independence: data should be collected using a simple random sample.
 - ii. For a test for homogeneity: data should be collected using a stratified random sample or randomized experiment.
 - iii. When sampling without replacement, check that $n \leq 10\% N$.b. Large counts (shape): all expected counts should be greater than 5.

Essential question: Why can identical counts require different chi-square procedure names and hypotheses?

DOK-3 skill: 4.C · **FRQ pattern:** design-controls-chi-square-procedure

DOK rationale: Part (c) coordinates sample structure, inferential target, hypotheses, shared expected counts, and conditions to adjudicate a procedure claim and explain why identical arithmetic does not imply identical population meaning.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask what information beyond the displayed counts names the procedure.

- Begin the 8.5 video follow-along at minute 5.
- Return at minute 33. Require a procedure and contextual hypotheses for each design.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one claim and cite how each design samples riders.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- In Design A, what population did each random sample come from?
- In Design B, which two variables were recorded on each sampled rider?
- What population statement changes when the procedure name changes?

Adult role

Solo teacher. Tie each procedure and hypothesis to its sampling structure, then compare the shared arithmetic.

[WARN] WATCH FOR

- Naming a test from the row labels without reading how the samples were taken.
- Using association language for homogeneity without identifying separate route populations.
- Claiming that different procedure names require different expected counts or degrees of freedom.

SCORING PART (c) — E / P / I

Expected elements

- **names-both-procedures** (required) — Names homogeneity for A and independence for B
- **states-distinct-hypotheses** (required) — States same-distribution and no-association hypotheses in context
- **uses-sample-structure** (required) — Uses three population SRSs versus one two-variable SRS
- **states-reader-consequence** (required) — Diagnoses the table-only rule's false sampling assumption
- **recognizes-shared-arithmetic** (required) — Identifies the shared numerical setup

E	Names both tests and hypotheses, uses sampling structure, diagnoses the rejected rule, and identifies shared arithmetic.
P	Distinguishes the tests but omits hypothesis context, the consequence, or shared arithmetic.
I	Chooses from row labels alone or says the designs need different expected counts.

[STAR] TODAY'S PROBLEM — annotated

Suppose a transit authority obtains the shown counts under either design.

Design A: Take independent SRSs of 100 from each of three route populations of 4,000 and record ticket method.

Design B: Take one SRS of 300 from 15,000 system riders and record both route and ticket method.

Ari: “Route rows imply homogeneity; the counts are enough to choose.”

Bea: “Read how riders were sampled; identical counts can support different population claims.”

Ticket method

Route	Mobile	Card	Cash	Total
North	60	25	15	100
East	48	32	20	100
West	42	33	25	100
Total	150	90	60	300

FIRST TAKE (5 min)

Which claim is more defensible? Cite the sampling structure of each design.

Key: Not graded. Sampling structure, not table shape, controls the choice.

Rules callout “LET DATA COLLECTION NAME THE TEST (keep on the page)” is printed on the student sheet.

DOK 1 (a) For each design, state the number of random samples and the categorical variable or variables recorded.

Key: Design A uses three independent SRSs and records one categorical response, ticket method. Design B uses one SRS and records two categorical variables, route and ticket method.

DOK 2 (b) Calculate the expected counts and df , then verify random sampling, 10 percent, and expected counts for both designs.

Key: Each row has expected counts (50, 30, 20), all greater than 5, and $df = (3 - 1)(3 - 1) = 4$. For A, the independent SRSs satisfy $100 \leq 0.10(4,000) = 400$ in each route population. For B, the SRS satisfies $300 \leq 0.10(15,000) = 1,500$.

DOK 3 (c) Adjudicate the claims. For each design, name the procedure and state H_0 and H_a in context. Use the sampling structure, diagnose the rejected table-only rule, and identify the numerical pieces that remain identical.

Key: Bea is warranted. For A, use homogeneity: H_0 says ticket-method distributions are the same across route populations; H_a says at least one differs. For B, use independence: H_0 says route and ticket method are not associated among system riders; H_a says they are associated. Three route-population SRSs versus one system SRS with two variables controls the choice. The rejected rule falsely assumes separate population samples for B. Observed and expected counts, $df = 4$, the statistic, and p-value match; their population meanings differ.

EXIT

One thing the video changed about my first take: _____